

Name: \_\_\_\_\_

Class: \_\_\_\_\_

# SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

## HSC ASSESSMENT TASK 3

JUNE 2010

### MATHEMATICS Extension 1

**Time Allowed:** 70 minutes

**Instructions:**

- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a **new page**.
- Standard integrals can be found on the last page.

| Question 1 | Question 2 | Question 3 | Question 4 | Question 5 | Question 6 | Total |
|------------|------------|------------|------------|------------|------------|-------|
| /9         | /8         | /9         | /8         | /8         | /8         | /50   |

### Question 1

- a) Write the exact value of :
- i)  $\cos^{-1}(\cos \frac{3\pi}{2})$  1
  - ii)  $\sin^{-1}0.4 + \cos^{-1}0.4$  1
- b) Simplify  $\frac{\sin(\pi-x)}{\sin(\frac{\pi}{2}-x)}$  2
- c) Evaluate  $\log_9 30$  correct to 2 decimal places. 1
- d) Solve, leaving answers in exact form:
- i)  $3^{x-1} = 7$  2
  - ii)  $\ln(x^2) + \ln x = 1$  2

### Question 2

- a) Evaluate  $\lim_{x \rightarrow 0} \frac{\frac{x}{3}}{\sin 2x}$  1
- b) Differentiate:
- i)  $\tan 2x$  1
  - ii)  $\ln(\frac{x}{x^2+3})$  2
  - iii)  $\operatorname{cosec}^2 x$  2
  - iv)  $\sin^{-1} 3x$  2

### Question 3

- a) Find:
- i)  $\int e^{4x} dx$  1
  - ii)  $\int \frac{1+2x}{x^2} dx$  2
  - iii)  $\int \frac{x}{1+2x^2} dx$  1

- b) Using the substitution  $u = \tan x$ , or otherwise, find  $\int \sec^2 x \tan^2 x \, dx$ . 2
- c) i) Find  $\frac{d}{dx}(xe^{2x})$  1
- ii) Hence or otherwise, find  $\int xe^{2x} \, dx$  2

#### Question 4

- a) On the same axes, sketch  $y = \sin x$  and  $y = \ln(\sin x)$  for  $0 \leq x \leq \pi$ . 2  
Clearly label key features.
- b) i) Express  $\sin^2 x$  in terms of  $\cos 2x$  1
- ii) The curve  $y = \sin 2x$ , for  $0 \leq x \leq \pi$ , is rotated about the  $x$  axis. 3  
Find the total volume generated.
- c) Evaluate  $\sin\left[\tan^{-1}\left(\frac{5}{4}\right)\right]$  in exact form. 2

#### Question 5

A function  $f$  is defined  $f(x) = x^2 - 2x$ .

- a) State the largest positive domain for  $f$  to have an inverse function  $f^{-1}$ . 1
- b) State the domain and range of  $f^{-1}$ . 2
- c) Sketch  $f$  and  $f^{-1}$  on the same axes for the domains and ranges above. 2  
Clearly show key points.
- d) Find the inverse function  $f^{-1}(x)$ . 2
- e) Find the value of  $x$  for which  $f(x) = f^{-1}(x)$ . 1

#### Question 6

- a) Express  $\cos^{-1}\left(\frac{1}{3}\right) + \cos^{-1}\left(\frac{1}{4}\right)$  in the form  $\cos^{-1}M$ . 2
- b) i) Write the domain and range for  $y = 3\sin^{-1}\left(\frac{x}{2}\right)$  2
- ii) Sketch the curve in i) 1
- iii) Evaluate  $\int_0^1 3\sin^{-1}\left(\frac{x}{2}\right) \, dx$ . 3  
Leave your answer in exact form.

# SOLUTIONS

① a) i)  $\cos^{-1} 0 = \frac{\pi}{2}$

ii)  $\frac{\pi}{2}$

b)  $\frac{\sin x}{\cos x} = \tan x$

c)  $\frac{\log 30}{\log 9} \doteq 1.55$

d)  $\log(3^{x-1}) = \log 7$   
 $(x-1) \log 3 = \log 7$   
 $x \log 3 - \log 3 = \log 7$   
 $x = \frac{\log 7 + \log 3}{\log 3}$

e)  $\log_e(x^3) = 1$

$\therefore x^3 = e^1$

$\therefore x = \sqrt[3]{e}$

② a)  $\lim_{x \rightarrow 0} \frac{2x}{\sin 2x} \times \frac{1}{6} = 1 \times \frac{1}{6}$   
 $= \frac{1}{6}$  ①

b) i)  $2 \sec^2 2x$  ①

ii)  $\log x - \log(x^2 + 3)$   
 $\frac{dy}{dx} = \frac{1}{x} - \frac{2x}{x^2 + 3}$  ②

iii)  $y = \frac{1}{\sin^2 x}$   
 $\frac{dy}{dx} = \frac{0 - 2 \sin x \cos x}{\sin^4 x}$

②  
 $= \frac{-2 \cos x}{\sin^3 x} \text{ or } -2 \cot x \operatorname{cosec}^2 x$

iv)  $\frac{dy}{dx} = \frac{1}{\sqrt{1-(3x)^2}} \times 3$   
 $= \frac{3}{\sqrt{1-9x^2}}$  ②

③ a) i)  $\frac{e^{4x}}{4} + c$

ii)  $\int \left( \frac{1}{x^2} + \frac{2x}{x^2} \right) dx$   
 $= \int (x^{-2} + \frac{2}{x}) dx$   
 $= -x^{-1} + 2 \log x + c$   
 $= -\frac{1}{x} + 2 \log x + c$

iii)  $\frac{1}{4} \int \frac{4x}{1+2x^2} dx$   
 $= \frac{1}{4} \log(1+2x^2) + c$

b)  $\int \sec^2 x \tan^2 x dx = \int \sec^2 x u^2 \frac{du}{\sec^2 x}$

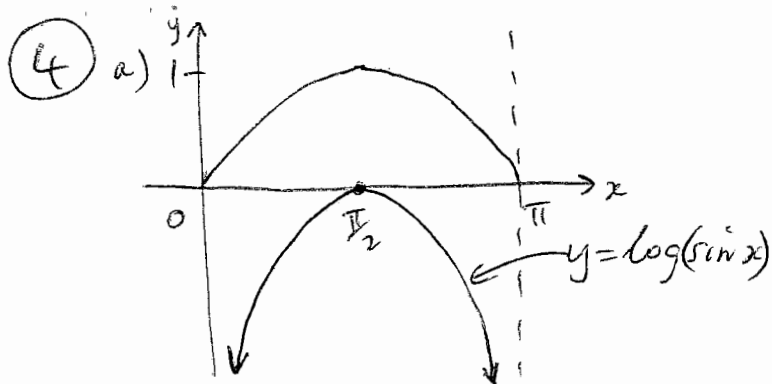
$u = \tan x$   
 $\frac{du}{dx} = \sec^2 x$   
 $dx = \frac{du}{\sec^2 x}$

$= \frac{u^3}{3} + c$   
 $= \frac{\tan^3 x}{3} + c$

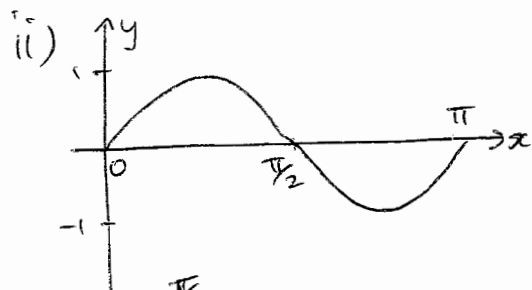
c) i)  $\frac{d}{dx} (x e^{2x}) = e^{2x} + 2x e^{2x}$

ii)  $2x e^{2x} = \frac{d}{dx} (x e^{2x}) - e^{2x}$

$\therefore \int x e^{2x} = \frac{1}{2} \int \frac{d}{dx} (x e^{2x}) - \frac{1}{2} \int e^{2x} dx$   
 $= \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + c$



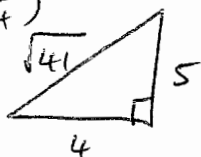
b) i)  $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$



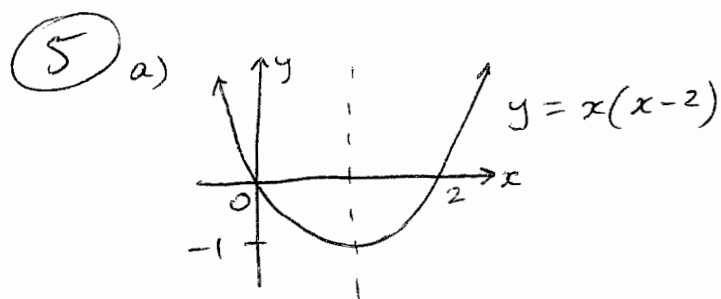
$$\begin{aligned} \text{Vol} &= 2\pi \int_0^{\pi/2} \sin^2 2x \, dx \\ &= 2\pi \int_0^{\pi/2} \frac{1}{2}(1 - \cos 4x) \, dx \\ &= \pi \left[ x - \frac{\sin 4x}{4} \right]_0^{\pi/2} \\ &= \pi \left[ \left( \frac{\pi}{2} - 0 \right) - (0 - 0) \right] \\ &= \frac{\pi^2}{2} \text{ units}^3 \end{aligned}$$

c) Let  $\alpha = \tan^{-1}\left(\frac{5}{4}\right)$

$\therefore \tan \alpha = \frac{5}{4}$

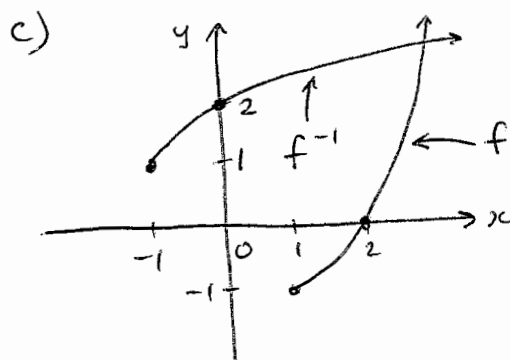


$$\begin{aligned} \therefore \sin\left(\tan^{-1}\frac{5}{4}\right) &= \sin \alpha \\ &= \frac{5}{\sqrt{41}} \end{aligned}$$



$\therefore$  domain:  $x \geq 1$

b) For  $f^{-1}$ ,  $D: x \geq -1$   
 $R: y \geq 1$



d)  $f^{-1}(x) \Rightarrow x = y^2 - 2y$

$$y^2 - 2y + 1 = x + 1$$

$$(y-1)^2 = x+1$$

$$y-1 = \pm\sqrt{x+1} \text{ only}$$

$$y = \sqrt{x+1} + 1$$

e) graphs intersect on  $y=x$

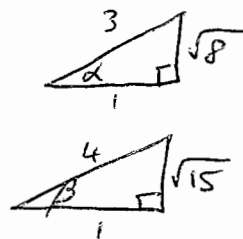
$\therefore$  solve  $x^2 - 2x = x$

$$x^2 - 3x = 0$$

$$x(x-3) = 0$$

$\therefore \underline{x=3 \text{ only}}$  ( $x=0$  not applicable).

⑥ a) Let  $\alpha = \cos^{-1}(\frac{1}{3}) \Rightarrow \cos \alpha = \frac{1}{3}$   
 $\beta = \cos^{-1}(\frac{1}{4}) \Rightarrow \cos \beta = \frac{1}{4}$



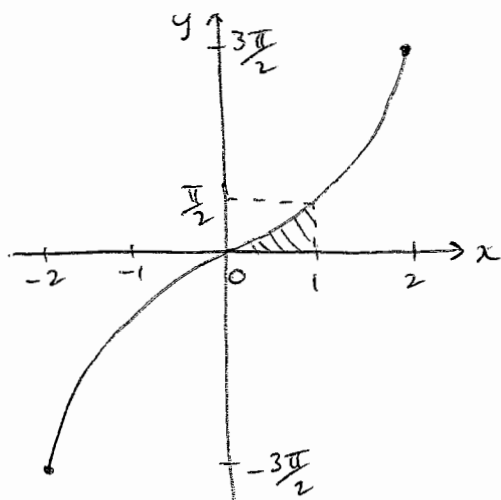
$$\begin{aligned}\cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \frac{1}{3} \times \frac{1}{4} - \frac{\sqrt{8}}{3} \times \frac{\sqrt{15}}{4} \\ &= \frac{1}{12} - \frac{\sqrt{120}}{12} \\ &= \frac{1 - \sqrt{120}}{12}\end{aligned}$$

$\therefore \alpha + \beta = \cos^{-1}\left(\frac{1 - \sqrt{120}}{12}\right)$  as reqd.

b) i) D :  $-2 \leq x \leq 2$

R :  $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

ii)



iii) Shaded area = rectangle -  $\int_0^{\pi/2} 2 \sin(\frac{y}{3}) dy$

$$\begin{aligned}\therefore \int_0^1 3 \sin^{-1}\left(\frac{x}{2}\right) dx &= \frac{\pi}{2} - \left[ -2 \cos\left(\frac{y}{3}\right) \times 3 \right]_0^{\pi/2} \\ &= \frac{\pi}{2} + 6 \left[ \cos \frac{y}{3} \right]_0^{\pi/2} \\ &= \frac{\pi}{2} + 6 \left( \frac{\sqrt{3}}{2} - 1 \right) \\ &= \frac{\pi}{2} + 3\sqrt{3} - 6\end{aligned}$$